SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1514

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I K.N. Rithreesh (192572097) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Cloud Computing

Definitions

Cloud Computing is a Model for delivering Computing Access such as Server, Storage, database, Networks and Software over the internet On a Pay as you go basis

Characteristics

- On demand Self Service
- Broad network access
- Resource Pooling
- Rapid elasticity
- Measured service

Benefits

- Cost efficiency
- Scalability & flexibility
- High availability & Reliability
- Disaster Recovery
- Access from anywhere

- Different clouds -
- Same sky -

Limitations

* Security & Privacy Concerns.
* Dependance on internet
* Potential for downtime
* Data migration challenges

Evolution of Cloud Computing

Hardware Evolution

* From mainframes (1960's-1970's) to Personal Computer (1980's)
* Rise Virtualization and Multicore-Processor (2001's)
* High Performance Servers, SSDs, GPU & Scalable

Distributed System

* From single to Machince system
* Peer to Peer grid computing
* Scalable

Service Delivery

* utility computing
* Emergence of Iaas, Paas, Saas
* Serreless & edge computing

Internet Software Evolution

* Static web Pages
* Dynamic web Application
* Cloud Native

3)

Service Model's.

IaaS

* Provide virtualized computing resource's over the internet
  Eg: virtual Machine
- Physical data centre
- Networking & storage Infrastructure
- Security & availability
* Maage os, Application and data

Paas

* Provides a complete Development & Runtime
* Infrastructure & hardware
* operating System
* Runtime environment and Middleware
* configure the Application System

  Eg: Azure

Saas

* Application software
* Data storage management
* User Access & Authentication
* Undate's, natches
* Manage own Data
- Usage Policies

  Eg: Salesforce.

XaaS

* Refer's to any cloud service Beyond the tradition Iaas, Paas & Saas.
* Provide Specialized cloud service
* Ensure security Performance & SLA
* Handle's Update's Scaling

Virtualization

- Enable's Multitenancy
- Provides on-Demand
- Reduce hardware cost
- Better Resource utilization
- Isolation & Security

web service

- Connects Application and Service across the cloud.
- Supports hybrid and Multi-Cloud environments

Eg: HTTP, REST

Cloud computing & virtualization

Enables Cloud Deployment

* Virtualization abstract hardware, allowing clouds Provider's to Run Multiple instance
* Together, they simplify setup, reduce complexity & speed up

Enable Scalabity

- Virtualization allows dynamic Provisioning and live Migration of virtual Machines.
- web service enables seamless Scaling through API's load Balance.

13/10/26